

INDIVIDUAL REFLECTION

Module 2: Guest Registration and Reservation Management

Working on Module 2: Guest Registration and Reservation Management of the Hotel Accommodation and Reservation Management System using C++ was a useful learning experience for me. This module helped me understand how Object-Oriented Programming concepts can be applied to manage important hotel operations such as guest registration, room availability checking, reservation creation, and reservation cancellation.

My main contribution was developing the guest registration and reservation management functions. I worked on storing guest details such as Guest ID, name, and phone number. I also implemented validation to ensure that duplicate Guest IDs were not registered. Before creating a reservation, the system checks whether the guest is registered and whether the selected room is available. This helped prevent invalid reservations and simple duplicate room bookings.

While developing the reservation module, I worked on connecting the Guest, Room, Reservation, and HotelSystem classes. A reservation is created by linking a registered guest with an available room and the number of nights required for the stay. The system generates a unique reservation ID and updates the room status from Available to Reserved. I also contributed to implementing reservation cancellation, where the room status is updated back to Available after a valid reservation is cancelled.

During the development process, I faced some challenges in connecting different classes and handling object relationships correctly. Understanding how pointers could be used to refer to room objects inside the reservation was initially difficult. I also faced challenges in checking room availability and ensuring that invalid guest IDs or reservation IDs were handled properly. Testing the program with different inputs helped identify errors and improve the validation logic.

This module helped me improve my understanding of important C++ concepts such as classes and objects, encapsulation, constructors, pointers, inheritance, member functions, vectors, conditional statements, and object interaction. I also gained practical experience in designing a menu-driven application and connecting multiple classes to perform a complete real-world operation.

The project is also related to Sustainable Development Goal 9 – Industry, Innovation and Infrastructure. A computerized hotel reservation system can reduce manual work and improve the organization of hotel operations. By digitally managing guest information and room reservations,

the system demonstrates how software technology can be used to improve efficiency and support innovation in service-based industries.

Overall, working on the Guest Registration and Reservation Management Module improved my C++ programming, debugging, and problem-solving skills. It helped me understand the importance of proper validation, class design, testing, and coordination between different modules. This assignment also contributed to achieving the mapped Course Outcome (CO) by helping me design, implement, test, debug, and evaluate a practical C++ application using Object-Oriented Programming concepts.